

BRYNJÓLFUR GAUTI GUÐRÚNAR JÓNSSON

PhD Student of Statistics at the University of Iceland

I am a PhD student of statistics at the University of Iceland, researching spatially distributed extreme rainfall and methods for modeling data-level dependence using copulas. Prior to this, I have worked extensively with Bayesian hierarchical models, having applied them to forecasting parliamentary elections and age-dependent mortality as well as the modeling of COVID-19 disease spread in Iceland during the pandemic. I also manage metill.is, an Icelandic website where I use official data to inform the discourse on current matters.



EDUCATION

- 2027 (est) • **University of Iceland**
PhD in Statistics
Reykjavík, Iceland
Subject: Improved 21st century projections of sub-daily extreme precipitation by spatio-temporal recalibration
- 2020 • **University of Iceland**
M.S. in Statistics
Reykjavík, Iceland
Thesis: [Continuous Categories and Age-Specific Mortality Rates: An application of Gaussian process priors to the Lee-Carter model](#)
- 2018 • **University of Iceland**
B.S. in Psychology
Reykjavík, Iceland
Thesis: [Up high, down low, too slow: Effects of nature on directed attention differ between high and low spatial frequencies](#). Results were presented at the Biomedical and Health Sciences Conference at the University of Iceland.

ACADEMIC EXPERIENCE

- 2024 • **StanCon 2024**
Oxford University
Oxford, England
• I presented my research and talked about modeling with copulas in Stan
[Presentation slides](#)
- 2024 • **Royal Statistical Society 2024 International Conference**
Doubletree Brighton Metropole
Brighton, England
• I gave a talk on my PhD research into developing Gaussian copulas for large spatial fields. [Presentation slides](#)
- 2022 | present • **PhD Student of Statistics**
University of Iceland
Reykjavík, Iceland
• Researching improved 21st century projections of sub-daily extreme precipitation and how hierarchical Bayesian spatio-temporal models can be applied to bias-correct simulations from convection-permitting climate models using observed meteorological data.

CONTACT INFO

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SKILLS

Experienced in data analysis and statistical learning models

Full experience with hierarchical Bayesian modelling.

Highly skilled in R, STAN, Python and LaTeX

Fully fluent in English and Norwegian

Experienced public speaker

Last updated on 2025-01-13.

- 2018 | 2022**
Statistician
 The Icelandic Heart Association 📍 Reykjavík, Iceland
 - High-dimensional statistical modeling of the effects of serum levels of proteins on heart failure and other outcomes.
- 2020 | 2020**
Statistician
 Directorate of Health 📍 Reykjavík, Iceland
 - Designed and implemented a Bayesian hierarchical generalized logistic growth model for short-term predictions of diagnosed cases at the start of the COVID19 pandemic.
- 2019**
Graduate Research Assistant
 The Science Institute, University of Iceland 📍 Reykjavík, Iceland
 - Performed preliminary research related to my MSc thesis and presented results at the 2019 Nordic Demographic Symposium in Reykjavík



PROFESSIONAL EXPERIENCE

- 2023 | 2024**
Data Scientist
 Lucinity 📍 Reykjavík, Iceland
 - Exploring the use of privacy-enhancing technologies, machine learning and network analysis for detecting money laundering activities
 - Graph neural networks, federated learning, simulating structured large-scale graph data
- 2018 | 2022**
Statistician
 The Icelandic Heart Association 📍 Reykjavík, Iceland
 - High-dimensional statistical modeling of the effects of serum levels of proteins on heart failure and other outcomes.
 - Diverse data analysis projects to help with publications
 - Writing software that helps coworkers with their reasearch
- 2018 | 2021**
Statistical Consultant
 The Statistical Consulting Center, School of Health Sciences, University of Iceland 📍 Reykjavík, Iceland
 - Assisted graduate students and academic researchers with data analysis and statistical testing
 - Did power calculations and helped plan methodology before data was gathered
 - Wrote methods chapters for research articles
- 2020 | 2020**
Statistician
 Directorate of Health 📍 Reykjavík, Iceland
 - Designed and implemented a Bayesian hierarchical generalized logistic growth model for short-term predictions of diagnosed cases at the start of the COVID19 pandemic.
 - High speed data analysis used to gather actionable intelligence presented via interpretable graphics, tables and text
 - [Designed and developed a detailed dashboard](#)
- 2018 | 2018**
Researcher
 Icelandic Center for Research - RANNÍS 📍 Reykjavík, Iceland
 - Designed and planned a research project and applied for funding
 - Developed personalized risk calculator to predict patients' risk of 30-day readmittance to psychiatric wards
 - One of five projects nominated for the President's Innovation Award
 - Results were presented at the Biomedical and Health Sciences Conference at the University of Iceland.



TEACHING EXPERIENCE

2020 present	● Statistical Consulting Teacher	📍 University of Iceland, Reykjavík, Iceland
2020 present	● Applied Data Analysis Teacher	📍 University of Iceland, Reykjavík, Iceland
2019 present	● R Programming Teacher	📍 University of Iceland, Reykjavík, Iceland
2019 2020	● Survival Analysis Teaching assistant	📍 University of Iceland, Reykjavík, Iceland